

Extended Abstract

Motivation A central goal in robotics is creating agents that can manipulate objects with human-like reliability, yet modern multi-fingered robots still cannot reliably grasp everyday objects in realistic settings. Existing methods are caught in a trade-off: open-loop generative models generalize to new objects but cannot react to errors during execution, while closed-loop reinforcement learning (RL) policies are more robust but fail when faced with novel environments. Mastering robust grasping is a cornerstone problem that would enable a cascade of downstream applications in automation, healthcare, and assistive care.

Method To address these challenges, we propose a framework that integrates efficient reinforcement learning with a novel reactive re-planning module. Our approach is built on three key components. First, we leverage Imitation Bootstrapped Reinforcement Learning (IBRL) to train a sample-efficient, vision-based student policy that learns to surpass the performance of its PPO-trained teacher. This is achieved by using the pre-trained imitation policy as a persistent reference to guide exploration and improve value estimation during online RL training. Second, we design a novel shaped reward function that combines sparse success signals with dense continuous rewards. Finally, our work is the first to introduce a hierarchical, heuristic-based reactive monitoring system that observes grasp execution, uses heuristics to detect failures, and initiates a new grasp when necessary. This enables high-level recovery without costly policy retraining.

Implementation All experiments are conducted within the Isaac Gym high-performance physics simulator. We use the ShadowHandGrasp task, where a 22-DoF Shadow Hand robot must grasp and lift objects from the UniDexGrasp dataset. Our process begins with collecting 50 successful demonstration trajectories from a pre-trained PPO teacher policy. These demonstrations are used to train an initial imitation policy via Behavioral Cloning (BC) for up to 100 epochs, using an AdamW optimizer and early stopping. The core of our experiment is the IBRL phase, which uses this BC policy to accelerate a TD3 agent. The actor and critic networks both use 3 hidden layers of 256 units. A replay buffer of 1,000,000 transitions is pre-filled with 10% expert data, and training batches maintain a 50% ratio of demonstration to online data.

Results Our IBRL-trained student policy achieved a 47.63% average success rate across attempt-level and object-level success for seen and unseen objects. A key finding emerged from our fine-tuning experiment: by fine-tuning the generalist IBRL policy on just a single object, its performance improved to 62.34%, significantly outperforming both the PPO teacher model (49.89%) and the DAgger-based student model from UniDexGrasp (38.73%). This surprising result suggests that localized fine-tuning can refine grasp behaviors in a way that generalizes across categories.

Discussion The quantitative results of our experiments are promising. The fact that our IBRL finetuned student policy surpasses both its expert teacher and the DAgger baseline confirms that our method provides a more effective learning signal, enabling the policy to discover superior strategies. In parallel, fine-tuning on a single object boosts generalization across diverse objects rather than causing overfitting. This suggests that the process acts as a form of targeted regularization. However, our work has limitations. All experiments are currently confined to simulation. Furthermore, our framework relies solely on vision and proprioceptive data, operating without the benefit of tactile sensing.

Conclusion In conclusion, this work presents a framework that integrates imitation-bootstrapped reinforcement learning with a reactive architecture to improve dexterous grasping. Our method yields a policy that is more sample-efficient and higher-performing than baseline approaches. We demonstrate that minimal fine-tuning can significantly enhance generalization, a key insight for future robotic learning. This research brings us closer to creating autonomous agents capable of robust, real-world manipulation, a cornerstone for advances in automation and human-robot collaboration. Future work will focus on scaling our experiments, adding richer sensory integration, and bridging the sim2real gap.

Reactive-Planning based Imitation-Bootstrapped Reinforcement Learning Enhances Dexterous Grasping Robustness and Generalizability with Minimal Finetuning

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Abstract

A central goal in robotics is creating agents that can reliably grasp diverse objects, a task hindered by a trade-off between the generalization of open-loop models and the reactive capabilities of closed-loop policies. To address this, we propose a framework that integrates efficient reinforcement learning with a novel reactive re-planning module. Our approach uses Imitation Bootstrapped Reinforcement Learning (IBRL) to train a student policy that surpasses its teacher, and is the first work to introduce a hierarchical monitoring system for in-execution failure recovery. Our policy based on IBRL achieves a 47.63% overall average success rate across 4 categories. We further demonstrate that fine-tuning this policy on just a single object provides a remarkable boost in performance to 62.34%, significantly outperforming the baselines, suggesting this acts as a form of targeted regularization that enhances generalization. This work brings us closer to truly reliable closed-loop grasping by combining a generalizable RL policy with an active re-planning mechanism.

1 Introduction

A central goal in robotics and artificial intelligence is to create agents capable of dexterous manipulation that mirrors the reliability and adaptability of human hands. Among these capabilities, robust dexterous grasping is arguably the most fundamental, underpinning a vast range of downstream applications. Despite decades of research, modern multi-fingered robotic hands still cannot reliably grasp diverse, everyday objects in realistic settings. The challenge is multifaceted, stemming from the high-dimensional complexity of multi-finger control, the immense variability in object properties, and the unpredictable nature of real-world environments Xu et al. (2023). These challenges have motivated interest in developing dexterous grasping systems that are both robust to execution-time disturbances and generalizable to unseen objects.

Mastering robust dexterous grasping would unlock a cascade of high-impact downstream applications. In industrial automation and warehouse logistics Laili et al. (2022); Solowjow et al. (2020), even a slight percentage boost in grasp success translates into substantial economic benefit. In operating

rooms, surgical robots that can reliably hold and manipulate delicate tissues promise less invasive procedures and faster patient recovery Xue et al. (2019). For in-home assistive care Smarr et al. (2012); Sawik et al. (2023), dependable grasping is the linchpin for helping the elderly or mobility-impaired. Beyond immediate economic and social benefits, advances in dexterous grasping push the broader frontier of safe, trustworthy human-robot collaboration, making it a cornerstone problem whose solution would ripple across the entire field of embodied AI.

The difficulty in achieving robust grasping has led to a split in dominant methodologies, each with significant limitations. On one hand, open-loop generative methods have demonstrated success in producing diverse and high-quality grasp poses offline, allowing them to generalize well across different objects and environments Weng et al. (2024); Lum et al. (2024a); Barad et al. (2024). However, their critical weakness lies in their inability to react to new information once execution begins. On the other hand, closed-loop reinforcement learning (RL) policies are inherently more robust to sensor noise and dynamic changes in a scene Kalashnikov et al. (2018); Herland et al. (2024). Yet, these systems often overfit to the limited set of objects and environments seen during training, causing their performance to collapse when confronted with novel scenarios. Recent hybrid and imitation learning approaches have attempted to bridge this gap. Some methods feed a single, high-quality grasp pose from a generative model into an RL policy Xu et al. (2023); Wan et al. (2023), while others use bootstrapping from imitation to improve sample efficiency Hu et al. (2023). While these are valuable steps forward, they lack the capacity to re-plan during execution.

To address these challenges, we propose a framework that integrates efficient reinforcement learning with a novel reactive re-planning module. Our approach is built on three key components. First, we leverage Iterative Bootstrapped Reinforcement Learning (IBRL) to train a sample-efficient, vision-based student policy that learns to surpass the performance of its PPO-trained teacher policy. Second, we design a novel shaped reward function that combines sparse success signals with dense rewards for proximity, contact, lifting, post-grasp stability, and time, guiding the policy toward more stable and effective behaviors. Finally, our work is the first to introduce a reactive monitoring system that observes grasp execution, detects failures, and initiates a new grasp when necessary, enabling high-level re-planning and recovery without costly policy retraining.

Our work aims to investigate: To what extent does integrating IBRL for policy learning improve sample efficiency, final grasp success rate, and the overall task robustness, compared with the UniDexGrasp baseline Xu et al. (2023)? How does incorporating a reactive planning module influence grasping behavior of dexterous hand? Our experiments yield several key findings. Our IBRL-trained student policy achieved a 47.63% average success rate across attempt-level and object-level success for seen and unseen objects. A key finding emerged from our fine-tuning experiment: by fine-tuning the generalist IBRL policy on just a single object, its performance improved to 62.34%, significantly outperforming both the PPO teacher model (49.89%) and the DAGger-based student model from UniDexGrasp (38.73%). This surprising result suggests that localized fine-tuning can refine grasp behaviors in a way that generalizes across categories. This work brings us closer to truly reliable closed-loop grasping by combining a generalizable RL policy with an active re-planning mechanism.

2 Related Work

2.1 Generative-Model-Based Approaches

Grasping research has largely progressed along two directions: generative approaches and learning-based approaches. The foundation of generative grasping was built on early generative models. Early probabilistic-model based approach Saxena et al. (2006) showed that data-driven synthesis could infer grasps from partial views. VAE-based approaches such as 6-DOF GraspNet Mousavian et al. (2019) and ZeroGrasp Iwase et al. (2025) then introduced a continuous latent grasp space that generalize for unseen objects. Concurrently, GAN-based approaches such as GraspGAN Bousmalis et al. (2018) were explored for their ability to refine grasp quality through an adversarial training process.

Building on these foundations, modern approaches have refined diffusion-based generative models to enhance their efficiency, versatility, and performance. DexDiffuser Weng et al. (2024) and Get-a-Grip Lum et al. (2024a) extends the diffusion concept to the challenging domain of high-dimensional dexterous grasping, generating precise multi-finger grasps. To address the computational cost of standard diffusion models, GraspLDM Barad et al. (2024) applies the diffusion process in a

compressed latent space rather than directly on the high-dimensional grasp parameters. In parallel, IGD Song et al. (2024) unifies dense affordance maps with sampling to output full grasp distributions, whereas RAGG Tang et al. (2025) conditions diffusion on similar object geometries, improving robustness further on novel shapes and unseen objects.

These generative methods are typically trained offline and tend to generalize well, particularly across diverse environments. However, their inherent design makes them open-loop and not reactive to disturbances during execution. In addition, the separation of sensing and control does not align with grasping behaviors of humans and animals. Kalashnikov et al. (2018) We use closed-loop RL-based design to overcome these limitations, enabling the robot to detect slippage and re-plan grasps on the fly.

2.2 Reinforcement-Learning-Based Approaches

In contrast to open-loop generative methods, Reinforcement Learning (RL) approaches train policies to perform the entire grasping pipeline in a closed-loop manner, from reaching to lifting. Foundational work like QT-Opt Kalashnikov et al. (2018) demonstrated that large-scale, self-supervised RL could learn effective grasping policies directly on physical robots. This end-to-end, vision-based paradigm has since matured, with modern methods learning continuous 6-DoF policies purely in simulation and transferring them zero-shot to the real world Herland et al. (2024). In parallel, some policies incorporate sensing modalities and leverage real-time force feedback to improve robustness and enable more delicate and precise interactions with objects Beddow et al. (2024).

Building on this reactive foundation, recent work has focused on overcoming RL’s primary challenges of sample efficiency and generalization. To address the issue of limited training diversity, GraspVLA Deng et al. (2025) pre-trains a foundational policy on a billion-scale synthetic dataset, enabling remarkable zero-shot generalization to novel objects and scenarios. To improve sample efficiency, OmniGrasp Luo et al. (2024) guide the difficult RL exploration process with learned priors such as a pre-trained humanoid motion model. Hybrid models such as UniDexGrasp Xu et al. (2023) and UniDexGrasp++ Wan et al. (2023) combines RL with generative approach, explicitly use a generative model to propose grasp configurations, which then serve as goals for a closed-loop RL policy. Similarly, DextrAH-G Lum et al. (2024b) combines RL with geometric fabric to address sim2real gap.

One common challenge across RL methods for dexterous grasping is their extreme training sample complexity. Our work address this by combining imitation learning and reinforcement learning to reduce sample complexity and speed up training.

2.3 Teacher-Student Distillation and Imitation-Learning-Based Approaches

Imitation Learning (IL) offers a straightforward paradigm for teaching robots complex dexterous grasping by learning directly from expert demonstrations. The aforementioned hybrid approaches uses teacher-student distillation to take advantage of IL’s efficient learning. Xu et al. (2023); Wan et al. (2023) In this framework, a student policy is trained to mimic the actions of an oracle teacher policy that has access to ground-truth environmental states not available in the real world. The student policy is then distilled from the teacher using the DAGger Ross et al. (2011) algorithm.

Classic behavioral cloning (BC), however, demands thousands of expert trajectories. Recent work therefore pivots toward few-shot regimes and improving training efficiency. Graph-alignment strategy augmented with in-context learning and energy-based model is able to generalize grasping from only 3-4 demonstrations Vosylius and Johns (2023). In parallel, vision-based IL approaches have also been explored, including the MIMO framework Cai et al. (2024) and CNN-based framework Zhang et al. (2024). Although these methods sharply cut data needs, they execute the derived trajectory open-loop. Another line of work augments scarce demonstrations. DemoGen Xue et al. (2025) warps a single human demo into diverse spatial configurations to synthetically populate a training set. Latent-diffusion planning in IL Xie et al. (2025) is capable of taking advantage of action-free demonstrations and suboptimal data. Complementary efforts either refine data quality or use auxiliary modules. Attaching colored grippers and apply RANSAC + ICP for precise, frame-level action labels enhance policy overall performance Song (2025). SKIL Wang et al. (2025) uses semantic key-points with vision foundation models to generalize across embodiments, whereas AdaDexGrasp Shi et al. (2025) couples a single human demo with RL fine-tuning and a vision-language selector.

However, most existing IL methods lack mechanisms to dynamically replan after failures or environmental changes, limiting real-world applicability. Our study builds on these insights, and is the first work to incorporate a reactive monitor that triggers re-proposal and recovery.

3 Method

3.1 High-Level Sketch

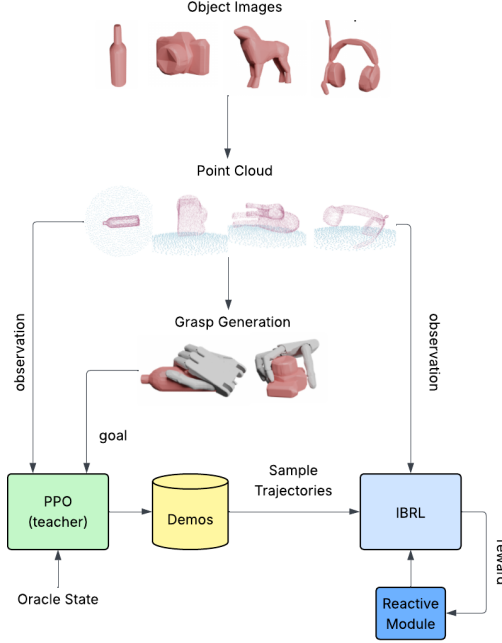


Figure 1: Diagram of our pipeline

Our proposed framework builds upon the structure and codebase of UniDexGrasp Xu et al. (2023). The process begins with an input of object vision images, which are converted into a point cloud. This point cloud serves two functions: it is used by a grasp generation module to create a task goal, and it acts as the primary observation for the learning agents. The teacher policy is a PPO agent, which uses the grasp goal along with privileged oracle state information to produce expert sample trajectories via a demo collector. The student policy is an IBRL agent that learns from these trajectories, but critically, it only uses the point cloud observation, not the oracle state. The student’s learning process is part of a closed loop, where its performance is refined by feedback from the novel reward function. The reactive module is activated in deployment loop, which allows the system to respond to events during execution.

3.2 Efficient Learning with Imitation Bootstrapped Reinforcement Learning (IBRL)

To achieve high sample efficiency and robust performance for dexterous grasping policy, we employ Imitation Bootstrapped Reinforcement Learning (IBRL) Hu et al. (2023) to train our student policy, a framework designed to effectively combine IL and online reinforcement learning RL. Imitation Learning demonstration data are taken from the PPO-trained teacher policy. The core principle of IBRL for grasping is to first train a standalone, high-capacity imitation policy, denoted as μ_{ψ} , on the available expert grasping demonstration data. By keeping the IL policy separate, we can leverage a much deeper and powerful network architectures that are well-suited for fitting grasping demonstration data but may be difficult to train in an online RL setting. This pre-trained IL policy is then used as a persistent reference to accelerate and stabilize the subsequent online RL training process through two key mechanisms: actor proposal and bootstrap proposal. This structure avoids the need for carefully tuned regularization losses to prevent the RL policy from deviating from the

expert’s behavior, as the system automatically balances between the two policies based on the learned Q-function.

During the online interaction phase, IBRL uses **actor proposal** to improve exploration. At each timestep t , both the evolving RL policy, π_θ , and the fixed imitation policy, μ_ψ , propose candidate grasping actions, a_t^{RL} and a_t^{IL} , respectively. The agent then executes the action that is estimated to yield a higher return by the current target Q-network, $Q_{\phi'}$. This selection process is formally expressed as:

$$a_t^* = \underset{a \in \{a_t^{\text{IL}}, a_t^{\text{RL}}\}}{\operatorname{argmax}} Q_{\phi'}(s_t, a)$$

This mechanism greatly accelerates exploration, especially in the early stages of training when the RL policy’s actions are often random and suboptimal, and rewards are unstable. By executing a high-quality action from the IL policy when the RL policy falters, the agent is more likely to encounter rewarding states, providing a stronger learning signal.

Concurrently, during the training phase, IBRL uses **bootstrap proposal** to enhance the quality of the target values used for updating the critic network. When computing the temporal difference (TD) target for a given transition, instead of bootstrapping solely from the action proposed by the target RL policy $\pi_{\theta'}$, IBRL considers actions from both the IL policy and the target RL policy in the next state, s_{t+1} . The TD target, y_t , is calculated by bootstrapping from the action that maximizes the future Q-value:

$$y_t \leftarrow r_t + \gamma \max_{a' \in \{a_{t+1}^{\text{IL}}, a_{t+1}^{\text{RL}}\}} Q_{\phi'}(s_{t+1}, a')$$

where $a_{t+1}^{\text{IL}} \sim \mu_\psi(s_{t+1})$ and $a_{t+1}^{\text{RL}} \sim \pi_{\theta'}(s_{t+1})$. This ensures that the Q-function is trained with more accurate and optimistic value estimates, which speeds up convergence and improves learning stability. By integrating the IL policy’s guidance into both the interaction and training loops, IBRL enables the student grasping policy to effectively leverage expert knowledge while retaining the flexibility to discover novel strategies and ultimately surpass the performance of the initial IL teacher.

Algorithm 1 Dexterous Grasping with IBRL

Require: Critics qty E , critic update qty G , policy update freq U , exploration std σ , noise clip c

- 1: Train imitation policy μ_ψ on expert trajectories from teacher policy $\mathcal{D}_{\text{grasp}} = \{\tau_1, \dots, \tau_n\}$
- 2: Initialize student RL policy π_θ , target policy $\pi_{\theta'}$, critics Q_{ϕ_i} , critics $Q_{\phi'_i}$ for $i = 1, \dots, E$
- 3: Initialize replay buffer $\mathcal{B}$ with teacher grasping trajectories $\mathcal{D}_{\text{grasp}}$
- 4: **for** $t = 1, \dots, \text{num_rl_steps}$ **do**
- 5: Observe state s_t from the grasping environment.
- 6: Compute IL action $a_t^{\text{IL}} \sim \mu_\psi(s_t)$ and RL action $a_t^{\text{RL}} = \pi_\theta(s_t) + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2)$.
- 7: Sample a set K of 2 indices from $\{1, 2, \dots, E\}$.
- 8: **Actor Proposal:** Take action a_t with higher Q-value
- 9: Store transition (s_t, a_t, r_t, s_{t+1}) in replay buffer $\mathcal{B}$
- 10: **if** $(t \bmod U) = 0$ **then**
- 11: **for** $g = 1, \dots, G$ **do**
- 12: Sample a minibatch of N transitions $(s^{(j)}, a^{(j)}, r^{(j)}, s_{+1}^{(j)})$ from $\mathcal{B}$
- 13: Sample a set K of 2 indices from $\{1, 2, \dots, E\}$
- 14: For each transition j in the minibatch, compute target Q-value:

$$a'^{\text{IL}} \sim \mu_\psi(s_{+1}^{(j)}) \quad \text{and} \quad a'^{\text{RL}} = \pi_{\theta'}(s_{+1}^{(j)}) + \text{clip}(\epsilon, -c, c)$$

$$\textbf{Bootstrap Proposal: } y^{(j)} = r^{(j)} + \gamma \max_{a' \in \{a'^{\text{IL}}, a'^{\text{RL}}\}} \left[\min_{k \in K} Q_{\phi'_k}(s_{+1}^{(j)}, a') \right]$$

- 15: Update critics ϕ_i by minimizing loss: $L(\phi_i) = \frac{1}{N} \sum_j (y^{(j)} - Q_{\phi_i}(s^{(j)}, a^{(j)}))^2$
 - 16: Update target critics $\phi'_i \leftarrow \rho \phi'_i + (1 - \rho) \phi_i$
 - 17: **end for**
 - 18: Update policy by maximizing $\frac{1}{N} \sum_j \min_{i=1, \dots, E} Q_{\phi_i}(s^{(j)}, \pi_\theta(s^{(j)}))$
 - 19: Update target actor $\theta' \leftarrow \rho \theta' + (1 - \rho) \theta$
 - 20: **end if**
 - 21: **end for**
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3.3 Shaped Reward Function for Stability

A key component of our training framework is a meticulously designed reward function that guides the agent toward stable and efficient grasping behaviors. To provide more granular feedback, we engineer a dense, multi-component reward function, R_t . This total reward at timestep t is formulated as:

$$R_t = w_{\text{succ}} r_{\text{success}} + w_{\text{prox}} p_{\text{prox}} + w_{\text{contact}} r_{\text{contact}} + w_{\text{lift}} r_{\text{lift}} - w_{\text{stable}} p_{\text{stable}} - w_{\text{time}} p_{\text{time}} - w_{\text{fall}} p_{\text{fall}}$$

This function includes several key terms: a proximity penalty, p_{prox} , based on the weighted L2 norm between the gripper (palm and fingertips) and the object’s handle; a contact reward, r_{contact} , which provides a bonus scaled by the distance to the goal once the fingertips are below a distance threshold from the handle; a lifting reward, r_{lift} , for raising the object above a height z_{lift} while maintaining contact; and a significant success bonus, r_{success} , when the object is within a tolerance ϵ_{succ} of its target. Finally, to encourage smooth and robust behaviors, we add penalties for large actions ($p_{\text{stable}} \propto \|a_t\|_2^2$), a large penalty p_{fall} if the object is dropped, and a small constant time cost p_{time} at each step. This structured guidance effectively decomposes the complex grasping task into manageable sub-goals, which dramatically accelerates convergence.

3.4 Reactive Monitoring and Re-planning

We are the first work to introduce a hierarchical system featuring a lightweight, reactive monitoring module, $\mathcal{M}$. This module operates concurrently with the low-level RL policy to oversee the grasp execution, using simple heuristics to detect failures like displaced objects. We implement two versions of this monitor. The first version, $\mathcal{M}_{GT}$, uses ground-truth object position, p_{obj} , from the simulator. It triggers a re-plan if object motion is detected before the grasp is initiated, i.e., if $\|\dot{p}_{obj}\| > \delta$. The second, more realistic version, $\mathcal{M}_{vision}$, employs a vision-based approach using GroundingDINO to track the object’s bounding box, $\mathcal{B}_{vision}$. A failure is flagged if the object exits the predefined operational workspace $\mathcal{W}$. Upon detecting a failure state $\mathcal{F}$, the monitor prompts a generative model, $\mathcal{G}$, to propose a new grasp pose, $g_{new} \leftarrow \mathcal{G}(s_t)$, which is then fed to the policy, and initializes a new grasp.

3.5 Simulation and Benchmarking

Our methodology is rigorously benchmarked across two primary axes: generalization and robustness. To assess generalization, we evaluate the policy’s performance on a comprehensive set of objects from the UniDexGrasp dataset, divided into seen (137 objects) and unseen (43 objects) categories. Success is measured using both attempt-level success rate (the percentage of individual grasp attempts that succeed) and object-level success rate (the percentage of objects successfully grasped after a set number of retries).

4 Experimental Setup

Our experimental setup is divided into three main stages: expert demonstration collection, behavioral cloning (BC) for policy initialization, and the final IBRL training phase. All experiments are conducted within the Isaac Gym simulation environment. We use the **ShadowHandGrasp** task within our custom UniDexGraspIBRL environment wrapper. In this state-based task, a 22-DoF Shadow Hand robot must grasp and lift various objects. The environment is configured to run 8 parallel instances for efficient data collection during RL training, with a maximum episode length of 200 steps.

4.1 Expert Demonstration Collection

To bootstrap our learning process, we first collect a dataset of 50 successful demonstrations. These are gathered from the PPO pre-trained teacher policy to ensure a high-quality dataset. During collection, we employ data augmentation, including adding minor noise (std: 0.01) to proprioceptive states, to enhance the robustness of the resulting policies. The final dataset is saved to `demonstrations.pkl` for use in subsequent stages.

4.2 Behavioral Cloning (BC)

The collected demonstrations are used to train an initial imitation policy via Behavioral Cloning. The BC network is a fully connected neural network with 3 hidden layers of 256 units each and a dropout rate of 0.1. The dataset is split into 80% for training and 20% for validation. We train the model for a maximum of 100 epochs using the AdamW optimizer with a learning rate of 3×10^{-4} and a weight decay of 1×10^{-4} . A cosine annealing learning rate scheduler is employed to gradually decrease the learning rate to a minimum of 1×10^{-6} . To prevent overfitting, we use early stopping with a patience of 15 epochs, monitoring the validation loss. The best-performing model, based on validation success rate, is saved as the BC policy for the next stage.

4.3 Imitation Bootstrapped Reinforcement Learning (IBRL)

The core of our experiment is the IBRL training phase, which leverages the pre-trained BC policy to accelerate a TD3 agent. The actor and critic networks both consist of 3 hidden layers with 256 units. The models are trained with the Adam optimizer and a learning rate of 3×10^{-4} . We use a replay buffer with a capacity of 1,000,000 transitions. Crucially, at the beginning of training, 10% of this buffer is pre-filled with the expert demonstrations from Stage 1. During training, batches of 256 transitions are sampled, with a 50% ratio (`bc_data_ratio: 0.5`) maintained between demonstration data and newly collected online data.

IBRL Algorithm Our IBRL implementation is built upon TD3, with γ set to 0.99 and τ to 0.005. Entropy is managed via automatic tuning, targeting a value of -22 (the negative of the action space dimension). The key IBRL mechanisms are enabled:

- **Q-Value Filtering** (`bc_q_filtering: true`): During action selection, the agent considers both the action proposed by the RL policy and the BC policy. It selects the action that yields a higher predicted Q-value, with the BC action’s Q-value being weighted by a factor of 0.5 (`bc_action_weight`).
- **Reactive Monitoring** (`enabled: true`): A safety mechanism is active to prevent catastrophic forgetting or policy collapse. If the RL policy’s running success rate over 100 episodes drops below 30% or its predicted Q-value falls below -2.0, the system temporarily switches to executing the deterministic BC policy to ensure stable data collection. This fallback is limited to a maximum of 20% of total training episodes.

Training and Evaluation The agent is trained for a total of 1,000,000 timesteps. Training commences after an initial 10,000 random exploration steps. The policy’s performance is evaluated every 10,000 timesteps by running 20 deterministic episodes. Checkpoints of the model are saved every 25,000 timesteps.

5 Results

Table 1: Grasping Performance Comparison Across Different Models

Method	Seen Category		Unseen Category	
	Attempt-Level	Object-Level	Attempt-Level	Object-Level
Teacher (PPO)	34.06 %	68.61 %	34.11 %	62.79 %
Student (DAGger)	26.76 %	58.39 %	23.26 %	46.51 %
Ours (IBRL)	41.36 %	69.34 %	25.58 %	54.26 %
Ours (IBRL, Finetuned)	53.53 %	79.56 %	46.51 %	69.77 %

5.1 Quantitative Evaluation

Our quantitative evaluation was designed to rigorously assess the performance of our proposed framework against established baselines across both seen and unseen object categories, with all results summarized in Table 1. The evaluation reveals not only the effectiveness of our core learning

algorithm but also the remarkable impact of our fine-tuning strategy. The baselines include a state-based Teacher model (PPO) adapted from UniDexGrasp Xu et al. (2023) and an accompanying vision-based Student model (Dagger). The teacher policy achieved a 34.06% attempt-level success rate on seen objects and a 34.11% rate on unseen objects, with object-level success rate 68.61% for seen objects and 62.79% for unseen. The DAgger student, expectedly, underperformed the teacher, with a success rate of only 26.76% on seen and 23.26% on unseen objects at attempt-level and 58.39% and 46.51% at object level, highlighting the inherent difficulty of mimicking a state-based expert from purely visual input through simple behavioral cloning.

Our primary student model, trained with Imitation Bootstrapped Reinforcement Learning (IBRL), demonstrated a significant performance improvement over both baselines. It achieved an attempt-level success rate of 41.36% on seen objects and an object-level rate of 69.34%, surpassing the PPO teacher from which it learned. Likewise, the success rates for unseen objects are also significantly strong, with 25.58% at attempt-level and 54.26% at object level. This superior performance can be attributed to the core mechanisms of IBRL. The actor proposal method improves exploration by allowing the agent to select the better of the actions proposed by the evolving RL policy and the fixed expert policy, preventing early-stage random exploration from derailing the learning process. Concurrently, the bootstrap proposal mechanism enhances the critic’s training by using a more accurate and optimistic target value derived from the best possible future action from either policy. This synergistic process enables the student policy to effectively leverage expert knowledge while retaining the flexibility to discover novel strategies that are more robust, ultimately allowing it to outperform the initial teacher policy.

The most compelling result of our study comes from the fine-tuned version of our IBRL model. By fine-tuning the generalist IBRL policy on just a single object, we observed a remarkable boost in performance across all categories. The attempt-level success rate on seen objects surged to 53.53%, and the object-level rate increased to 79.56%. Even more impressively, its performance on unseen objects also saw a dramatic increase, rising to 46.51% at the attempt level and 69.77% at the object level. We hypothesize that this counter-intuitive result occurs because fine-tuning on a single object acts as a form of targeted regularization. Instead of overfitting to that object’s specific geometry, the intense training forces the policy to learn more fundamental and generalizable physical principles of stable grasping—such as precise finger gaiting, contact point distribution, and force application. This refines the policy’s low-level motor skills in a way that transfers broadly, enhancing its ability to handle the complexities of both familiar and novel objects.

5.2 Qualitative Analysis

While quantitative metrics provide a clear measure of success, a qualitative analysis of our models’ behaviors offers deeper insights into *why* they succeed or fail, revealing the underlying strategies learned. By observing simulation rollouts, we analyzed the quality of grasp poses, the stability of lifts, and the emergent behaviors of our different policies. This analysis confirms that the performance gains seen in our results table correspond to qualitatively more robust and physically intelligent manipulation strategies, particularly highlighting the stability of the IBRL agent and the crucial role of our reactive-planning module in recovering from failure.

Our investigation began by comparing the grasping behaviors of the baseline models with our IBRL-trained agent. The Student model (Dagger) frequently exhibited brittle and hesitant behaviors. For instance, when attempting to grasp slender or irregularly shaped objects, the DAgger policy would often approach with a misaligned pre-grasp shape, causing it to knock the object over or make insufficient contact, leading to failure upon lifting. In stark contrast, our student model (IBRL) demonstrated consistently more stable and deliberate actions. When approaching the same objects, the IBRL policy would pre-shape its fingers to conform to the object’s surface before making contact, resulting in a secure, multi-point contact grip. This superior behavior can be directly attributed to our methodology; the shaped reward function explicitly guides the agent to value stable contact and lifting phases, while the IBRL framework’s exploration allows the student to discover these robust strategies that are more effective than simply mimicking the teacher’s "perfect" but potentially brittle trajectories. The agent learns not just a target pose, but a robust process for achieving a secure grasp.

The qualitative difference was even more pronounced in the IBRL finetuned on one object model. This policy displayed emergent, seemingly intelligent behaviors that were not explicitly programmed. For example, when faced with a challenging object like a irregular prism lying flat, the

finetuned agent would not attempt a direct power grasp. Instead, we observed it using one or two fingers to slightly nudge and lift the prism, reorienting the object into a more favorable position for a stable grasp before closing the remaining fingers to secure it. This behavior strongly supports our hypothesis that single-object fine-tuning acts as a form of targeted regularization. The intense training on one object’s complex dynamics forces the policy to master fundamental physical interactions, which it then creatively applies to solve grasping challenges with novel objects, leading to its superior performance on the unseen object category.

Finally, the effectiveness of the **reactive-planning module** was observed during policy deployment within the dynamic Isaac Gym environment. In representative scenarios where a stochastic event would displace the target object mid-execution, the module’s utility became clear. For instance, as the robotic hand descends to grasp a mug, a simulated external force displaces it laterally. Without our module, the policy would continue its original trajectory, closing its fingers on empty space and failing the attempt. However, with our system enabled, the state-based monitor, $\mathcal{M}_{GT}$, which uses ground-truth object position from the simulator, immediately detects that significant object motion has occurred, triggering a failure state. Upon detection, the high-level monitor terminates the failed attempt and directly initiates a restart of the entire grasping sequence. A new grasp attempt begins, using a new goal pose proposed by the generative model, $\mathcal{G}$, based on the object’s current location. This scenario vividly illustrates the critical advantage of our hierarchical approach: it decouples failure detection and recovery from the low-level control policy, enabling systematic and automated retries without requiring complex mid-execution replanning.

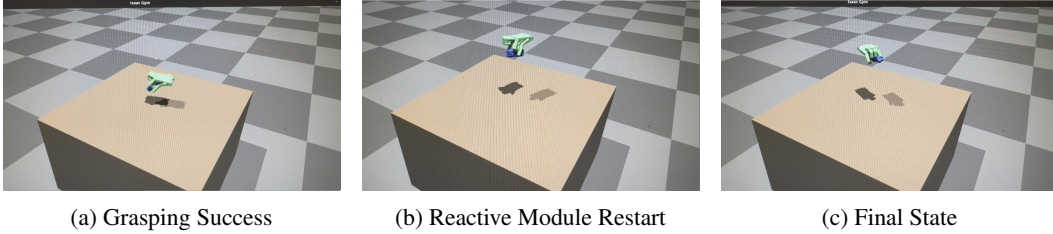


Figure 2: Grasping in Isaac Gym

6 Discussion

Our findings demonstrate a promising step toward more robust and generalizable dexterous grasping. The success of the IBRL-trained policy, especially the significant performance gain from minimal fine-tuning, suggests that our integrated approach can effectively bridge the gap between sample-efficient learning and high-performing, generalist policies. The proposed reactive monitoring system, based on success in qualitative analysis, has the potential to address the critical issue of in-execution failure recovery, a major roadblock for real-world robotic applications. Reflecting on the broader impact, advancements in robust dexterous manipulation stand to unlock significant societal and economic benefits. This research contributes to the foundational technology for safer and more trustworthy human-robot collaboration. However, our study has several limitations that must be acknowledged and also faced a number of challenges during its execution. A primary limitation of our current work is that all experiments were conducted exclusively in simulation. While Isaac Gym provides a high-fidelity physics environment, the sim-to-real gap remains a significant hurdle. Therefore, the true efficacy of our approach awaits validation on a real robotic platform. Furthermore, a second significant limitation is the framework’s reliance solely on vision and proprioceptive data, operating without the benefit of tactile sensing. This absence of multimodality prevents the agent from detecting critical physical phenomena such as contact forces and incipient slip.

Throughout this project, we encountered several technical difficulties. The initial setup and debugging of the Isaac Gym environment with the custom ShadowHand asset proved to be a non-trivial engineering challenge. A significant portion of our effort was dedicated to designing and tuning the shaped reward function; balancing the various reward components to guide the agent toward successful grasps without creating unforeseen behavioral artifacts required extensive experimentation. Furthermore, the integration of the IBRL algorithm was complex and computationally intensive, given the interplay between the pre-trained BC policy and the online TD3 agent.

7 Conclusion

In this work, we introduced a novel framework that enhances dexterous grasping by integrating a sample-efficient learning algorithm with a reactive re-planning module. Our approach addresses the critical limitations of existing systems: the inability of open-loop methods to react to disturbances and the poor generalization of many closed-loop policies. By leveraging IBRL, a carefully shaped reward function, and a hierarchical monitoring system, we have demonstrated a significant step toward creating robotic agents capable of robust and generalizable manipulation that mirrors the adaptability of human hands. Our results show that IBRL enables stronger policies, as our student policy trained with IBRL significantly outperforms both the PPO teacher and DAgger baseline. We also found that small-scale finetuning has a big impact; finetuning on just one object significantly improves generalization across diverse objects. Ultimately, our work is the first to synthesize reactive monitoring and generalizable reinforcement learning to facilitate more robust and reliable dexterous manipulation in dynamic environments. Future could focus on several key areas, beginning with sim-to-real transfer to bridge the gap from simulation to physical robotic systems by addressing challenges like sensor noise, calibration, and real-world dynamics. Extending the framework to advanced manipulation skills beyond grasping, such as in-hand object reorientation or pre-grasp manipulation is also a promising direction. Finally, progress can be made through richer sensory integration, particularly by incorporating tactile sensing to enhance grasp stability, object recognition, and reactive capabilities.

8 Team Contributions

- **Yi Du** lead project design and innovation, including integrating generative proposals with closed-loop RL and reactive re-planning.
- **YuCheng Yuan** implemented the calling structure and details of reward function, integrated into training pipeline, tuned hyperparameters, analyze result, and draft report.
- **Ryan Rong** integrated IBRL with the grasping task and wrote the demo collection, training, and evaluation pipelines. He helped run experiments and collected and analyzed results.

Changes from Proposal Not much, with necessary intimate co-collaboration in final phase

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